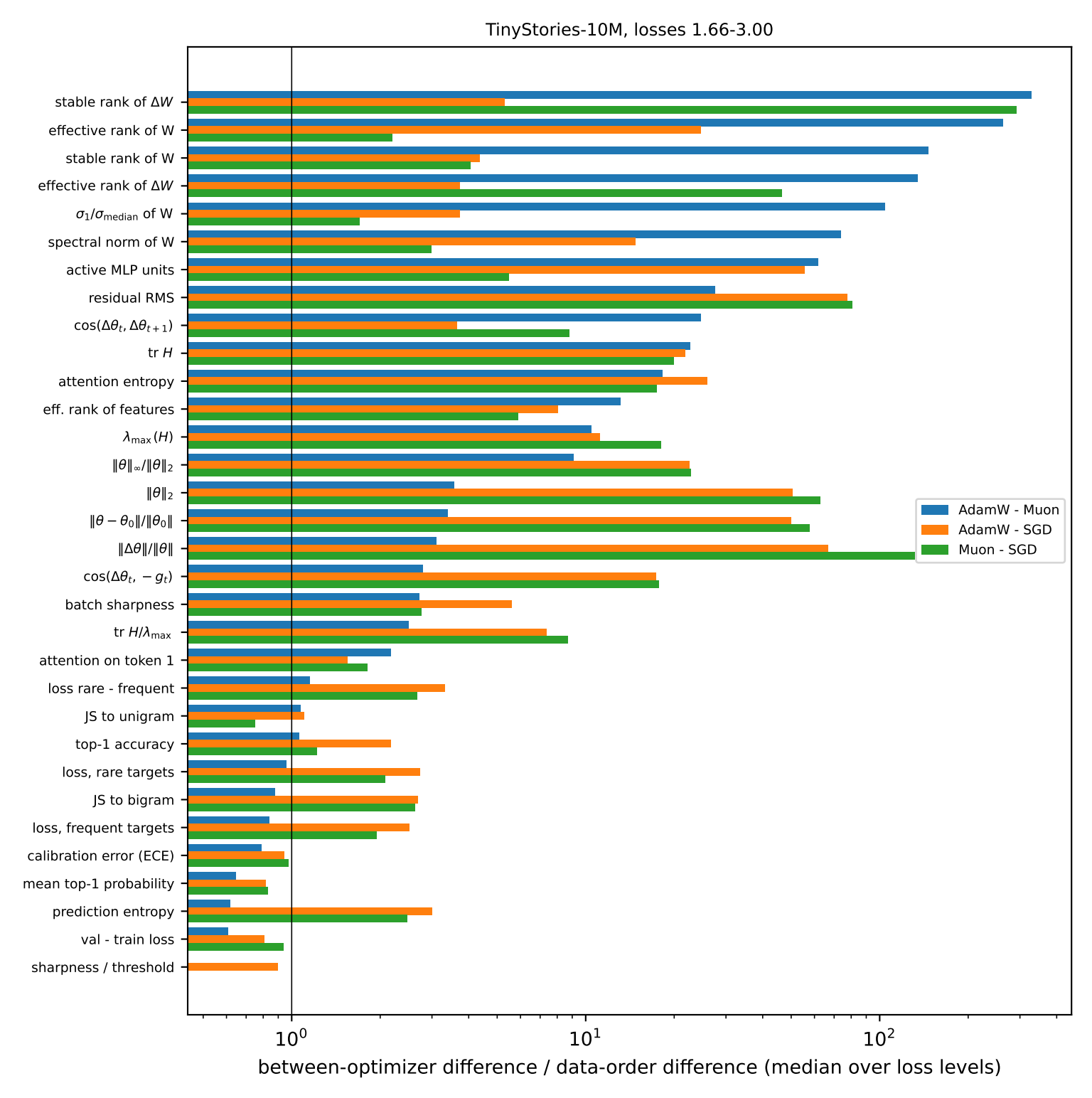
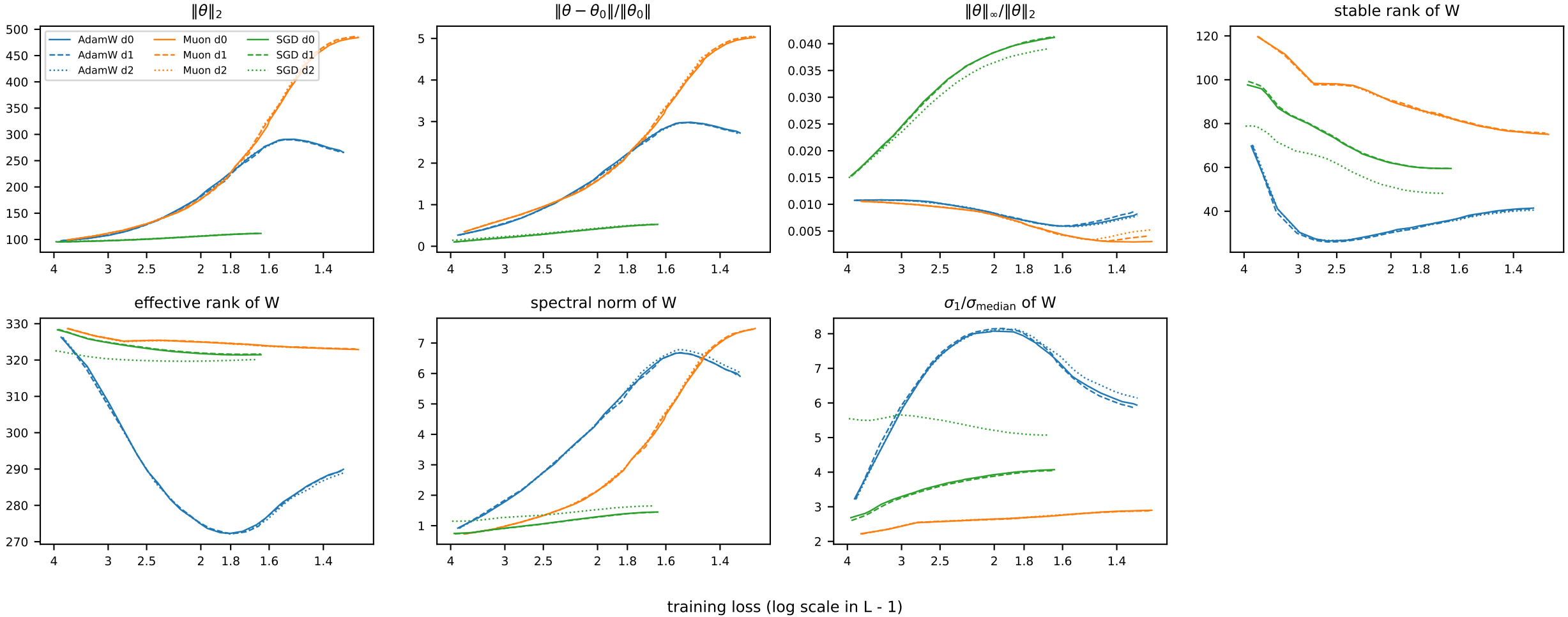
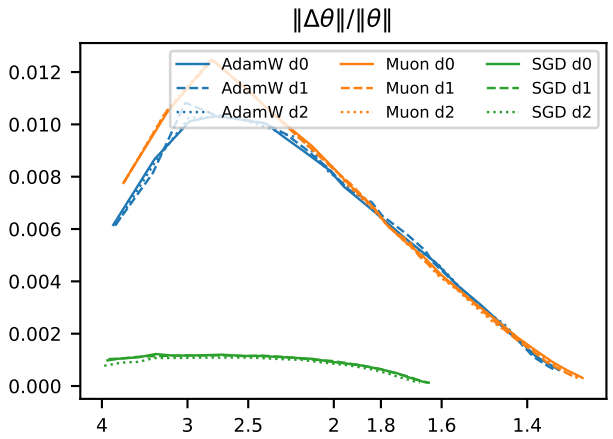
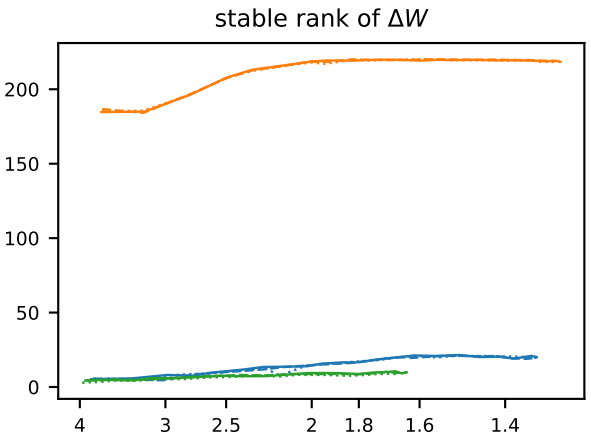
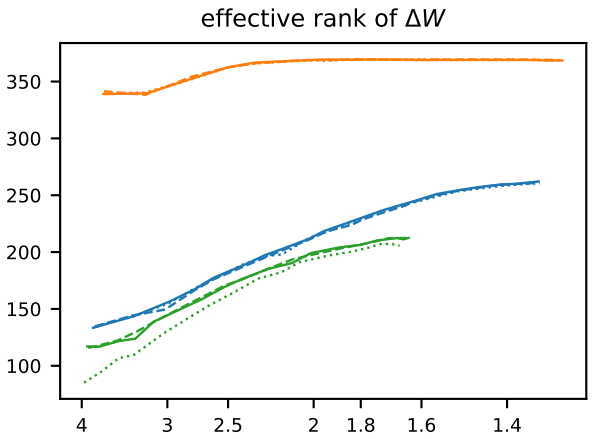
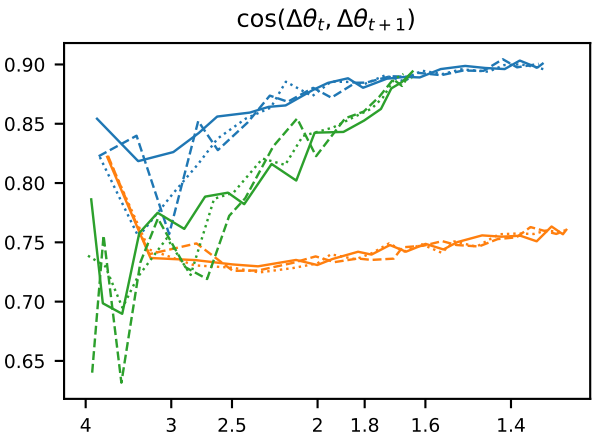

TinyStories-10M, losses 1.66-3.00
stable rank of ΔW
effective rank of W
stable rank of W
effective rank of ΔW
σ1/σmedian of W
spectral norm of W
active MLP units
residual RMS
cos(Δθt, Δθt+1)
tr H
attention entropy
eff. rank of features
λmax(H)
‖θ‖∞/‖θ‖2
‖θ‖2
‖θ − θ0‖/‖θ0‖
‖Δθ‖/‖θ‖
cos(Δθt, −gt)
batch sharpness
tr H/λmax
attention on token 1
loss rare - frequent
JS to unigram
top-1 accuracy
loss, rare targets
JS to bigram
loss, frequent targets
calibration error (ECE)
mean top-1 probability
prediction entropy
val - train loss
sharpness / threshold
AdamW - Muon
AdamW - SGD
Muon - SGD
10^0
10^1
10^2
between-optimizer difference / data-order difference (median over loss levels)

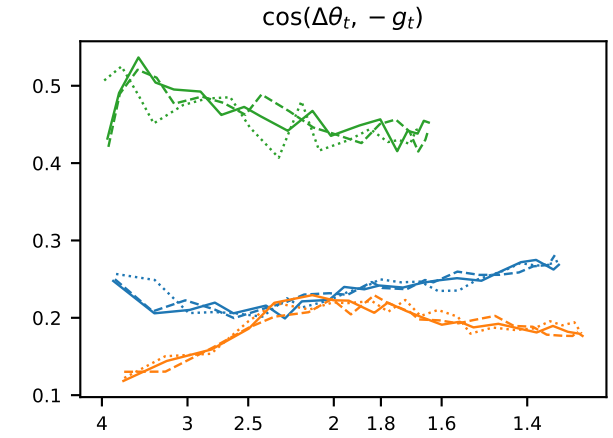

$\|\theta\|_2$
AdamW d0
AdamW d1
AdamW d2
Muon d0
Muon d1
Muon d2
SGD d0
SGD d1
SGD d2
$\|\theta - \theta_0\|/\|\theta_0\|$
$\|\theta\|_\infty/\|\theta\|_2$
stable rank of W
effective rank of W
spectral norm of W
$\sigma_1/\sigma_{\text{median}}$ of W
training loss (log scale in L - 1)



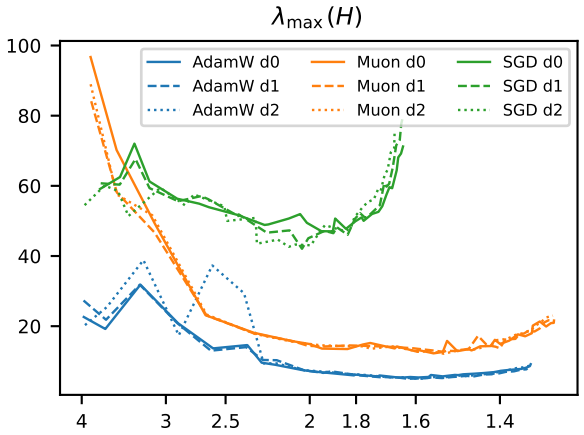


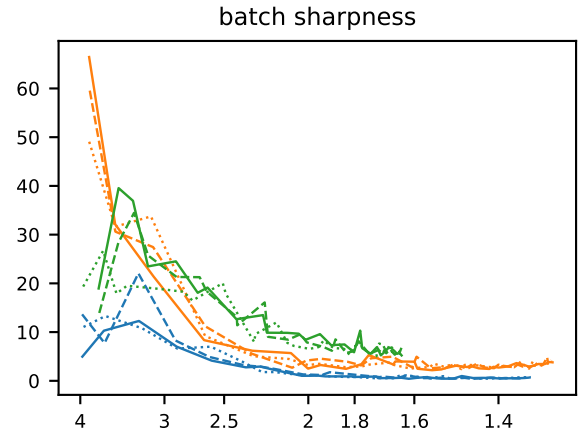


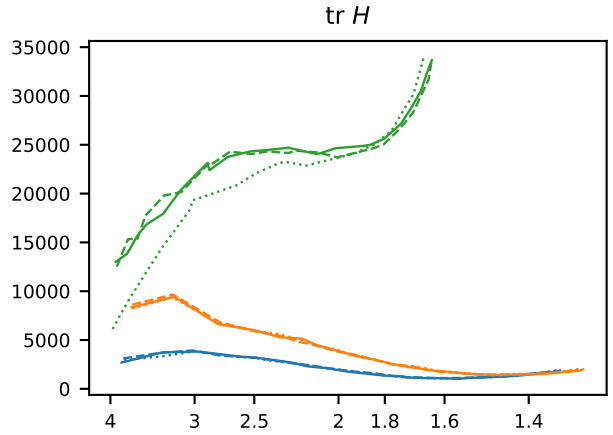


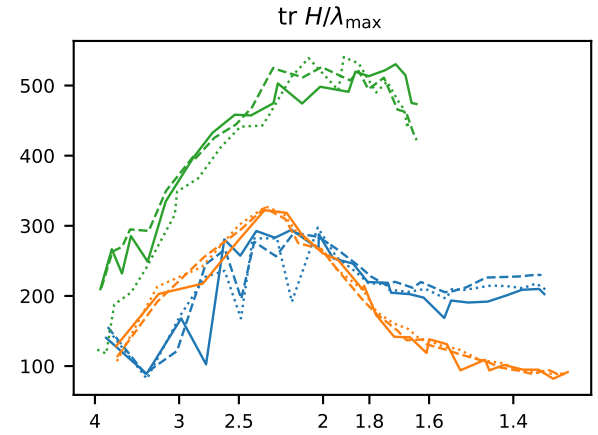


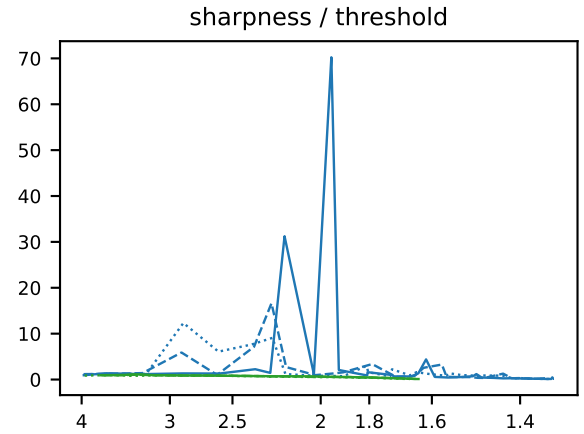
training loss (log scale in L - 1)



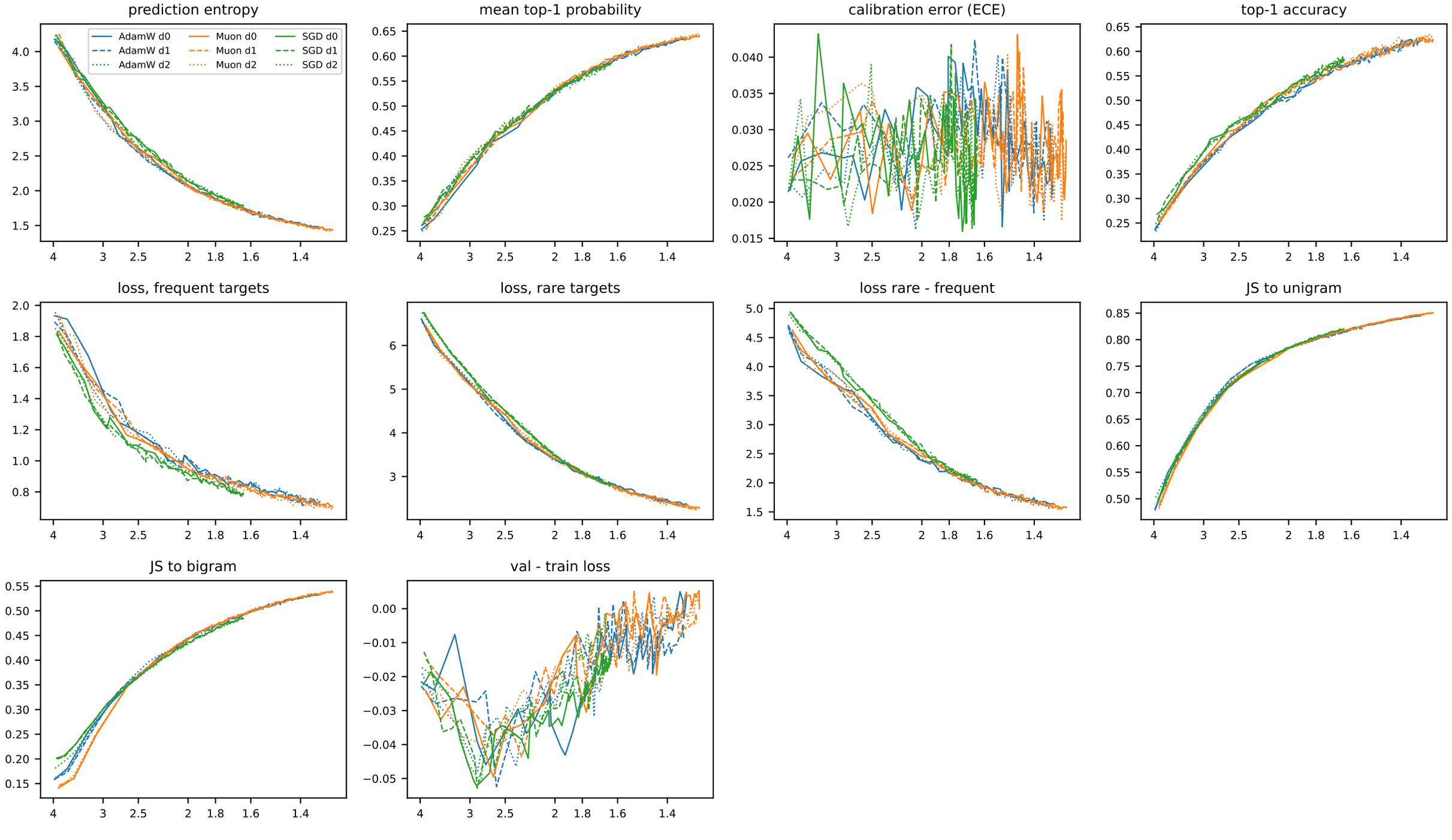


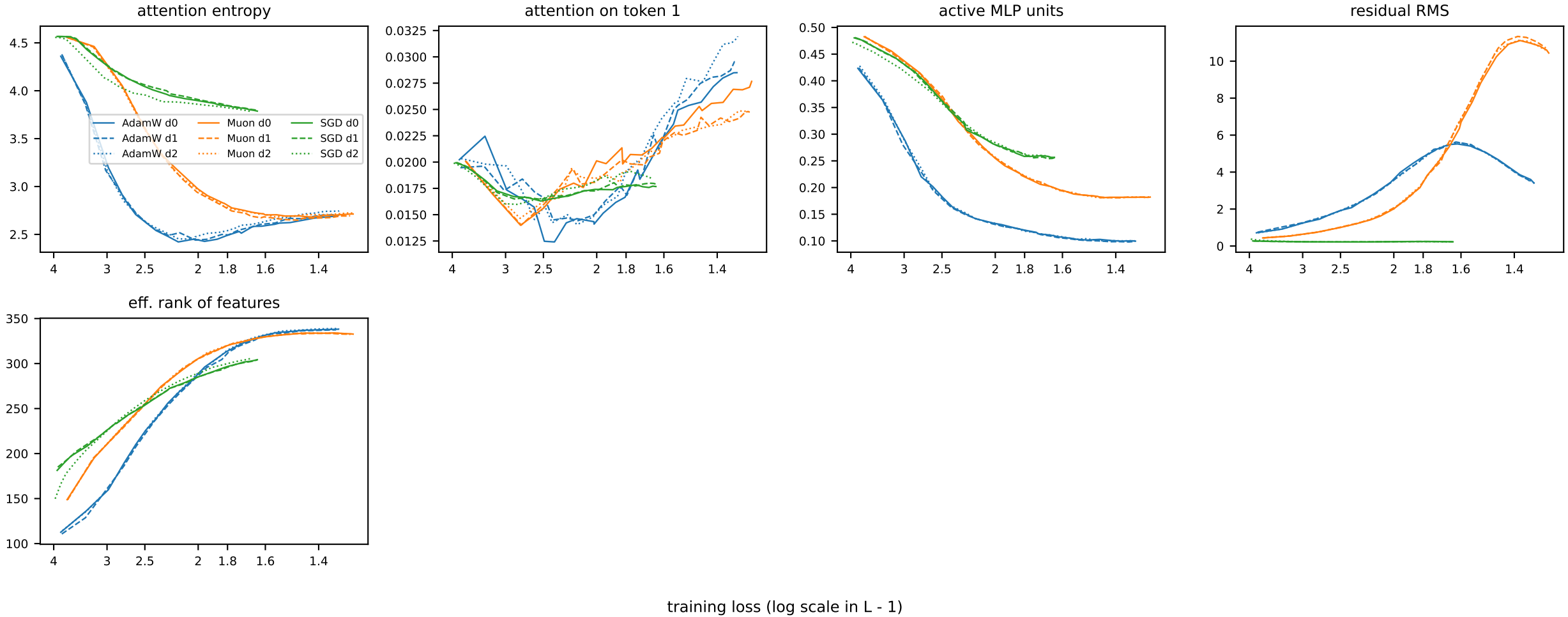




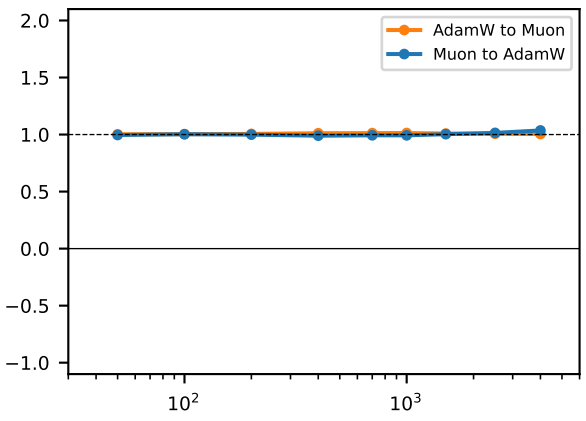


training loss (log scale in L - 1)

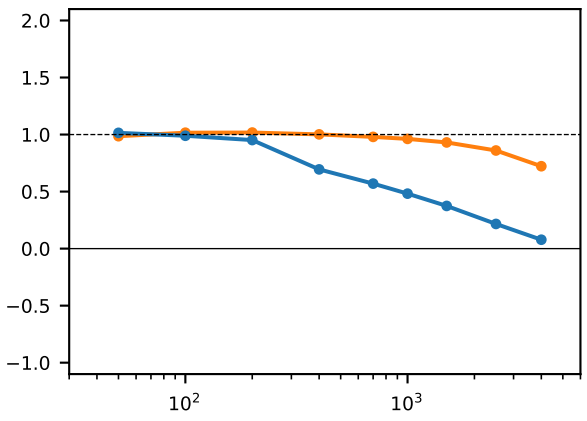

prediction entropy
mean top-1 probability
calibration error (ECE)
top-1 accuracy
loss, frequent targets
loss, rare targets
loss rare - frequent
JS to unigram
JS to bigram
val - train loss
AdamW d0
AdamW d1
AdamW d2
Muon d0
Muon d1
Muon d2
SGD d0
SGD d1
SGD d2
training loss (log scale in L - 1)


attention entropy
attention on token 1
active MLP units
residual RMS
eff. rank of features
AdamW d0
AdamW d1
AdamW d2
Muon d0
Muon d1
Muon d2
SGD d0
SGD d1
SGD d2
training loss (log scale in L - 1)

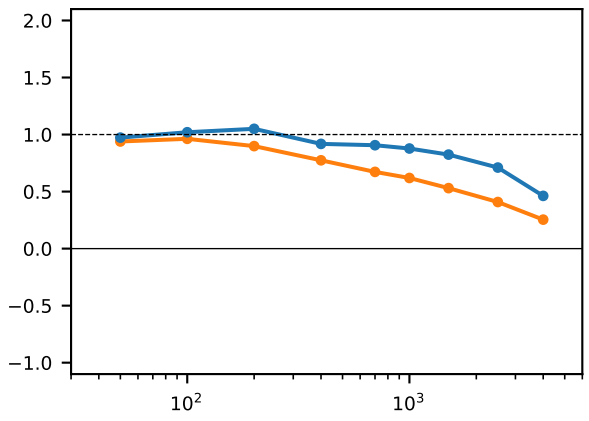
stable rank of ΔW (gap 401.5 floors)



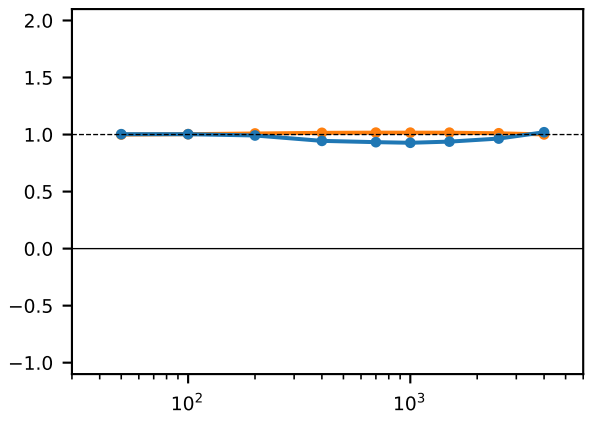
effective rank of W (gap 51.7 floors)



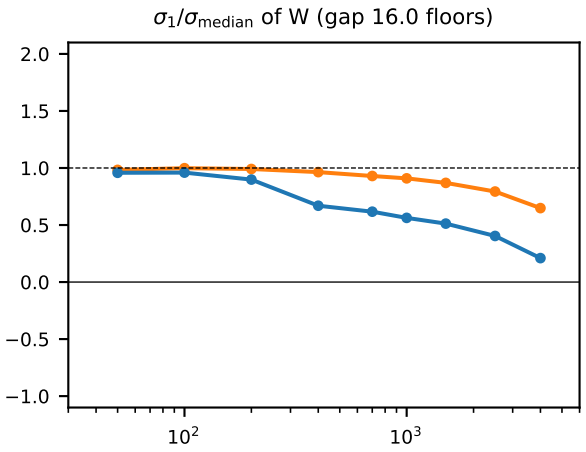
stable rank of W (gap 55.3 floors)



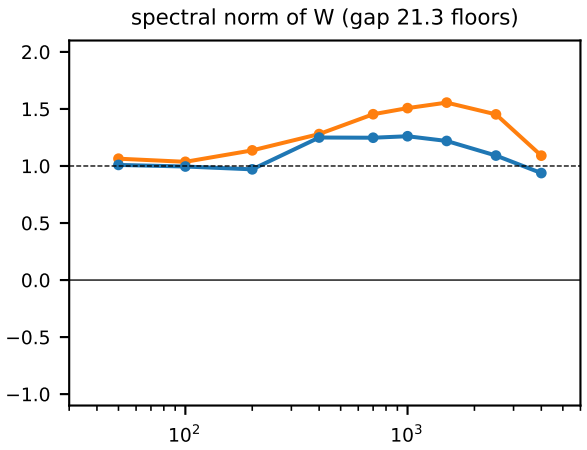
effective rank of ΔW (gap 95.6 floors)



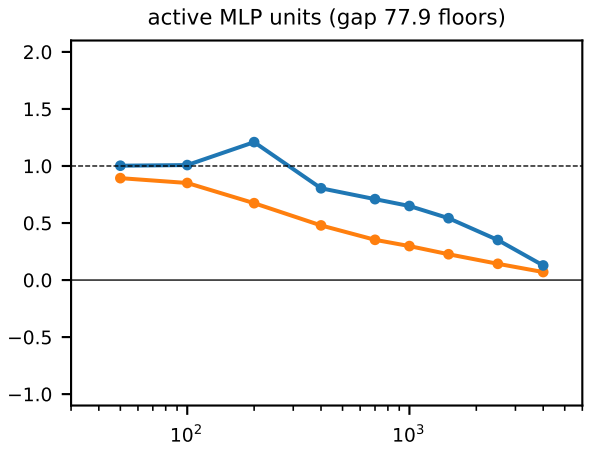
$\sigma_1/\sigma_{\text{median}}$ of W (gap 16.0 floors)



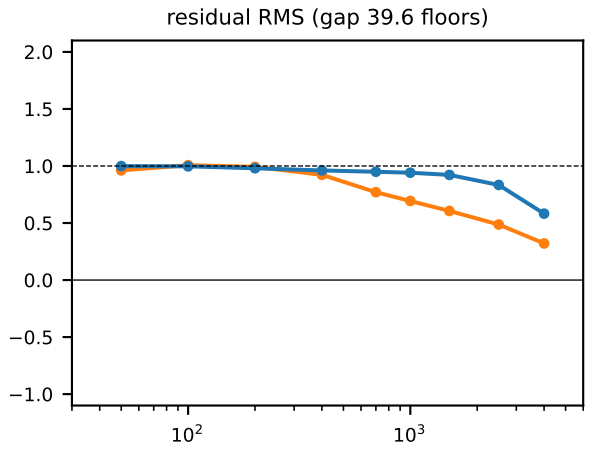
spectral norm of W (gap 21.3 floors)



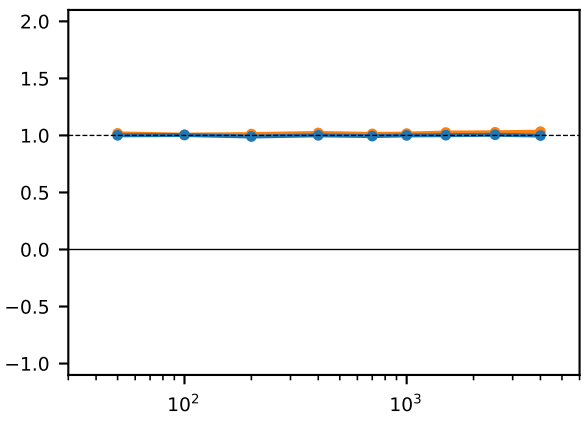
active MLP units (gap 77.9 floors)



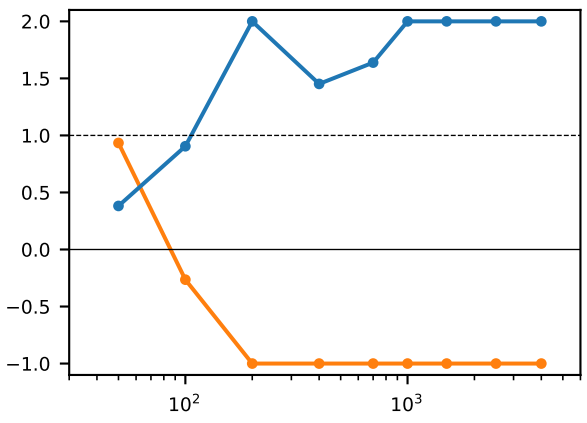
residual RMS (gap 39.6 floors)



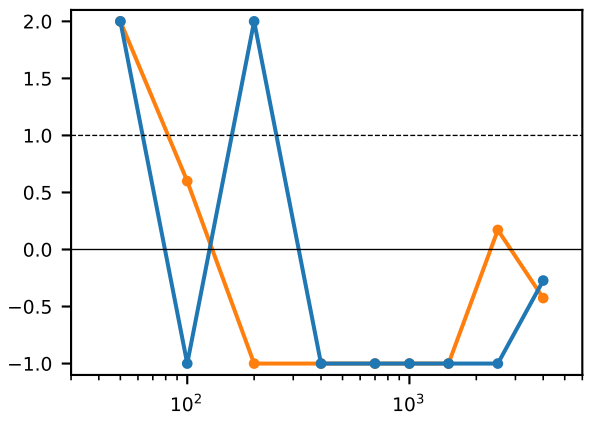
$\cos(\Delta\theta_t, \Delta\theta_{t+1})$ (gap 44.8 floors)



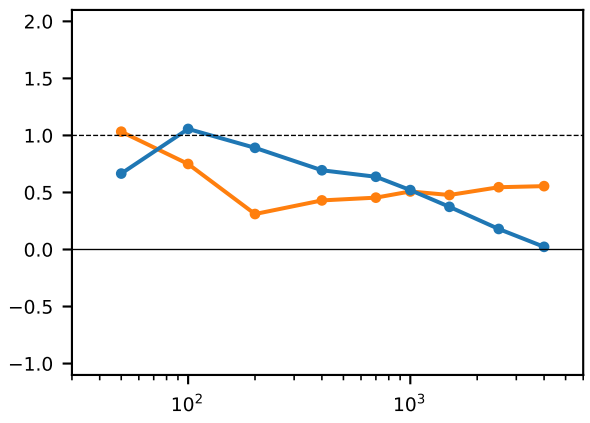
tr H (gap 1.6 floors)



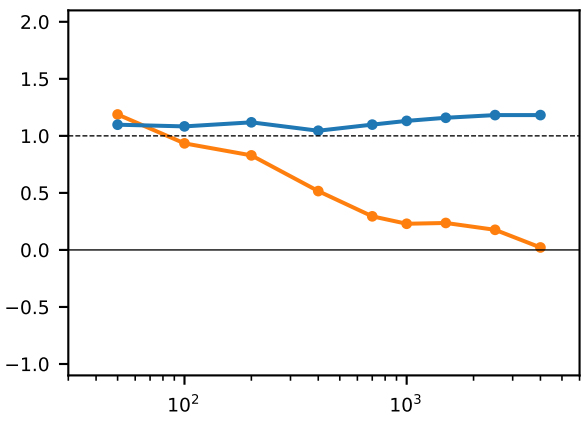
attention entropy (gap 0.4 floors)



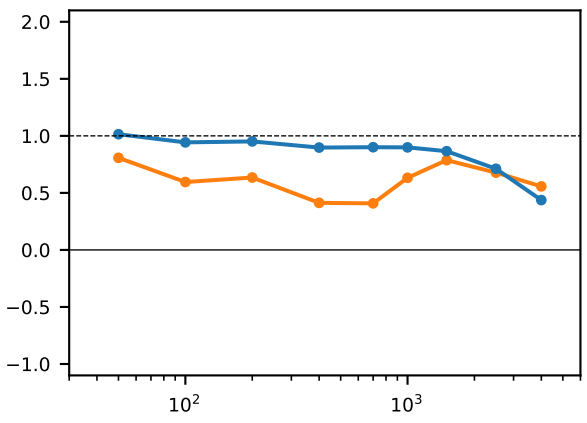
eff. rank of features (gap 5.2 floors)



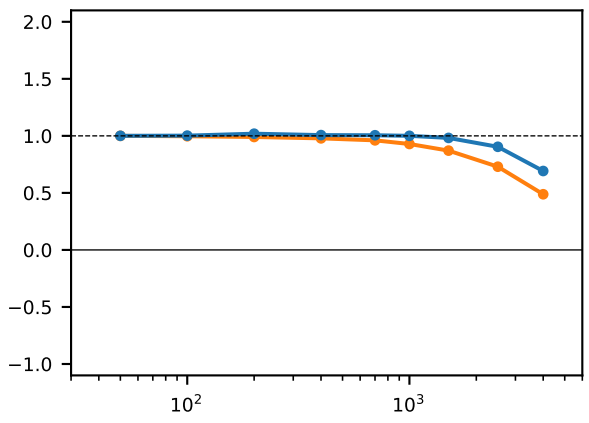
$\lambda_{\max}(H)$ (gap 13.4 floors)



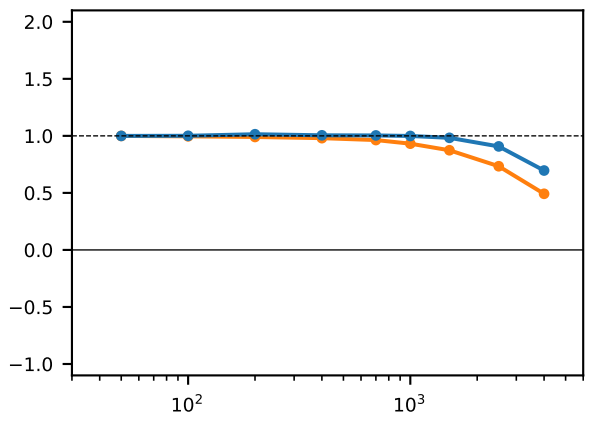
$\|\theta\|_\infty/\|\theta\|_2$ (gap 3.5 floors)



$\|\theta\|_2$ (gap 179.1 floors)

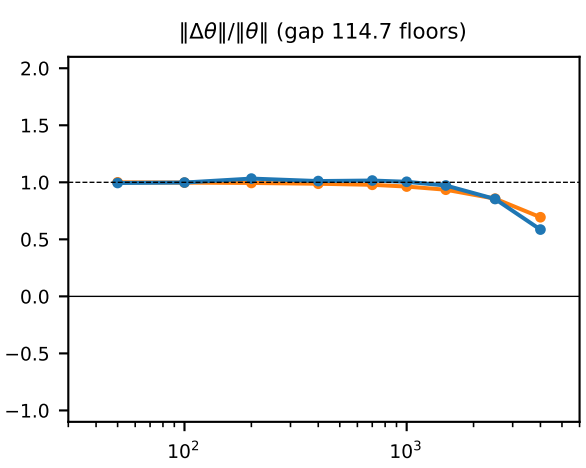


$\|\theta - \theta_0\|/\|\theta_0\|$ (gap 174.0 floors)

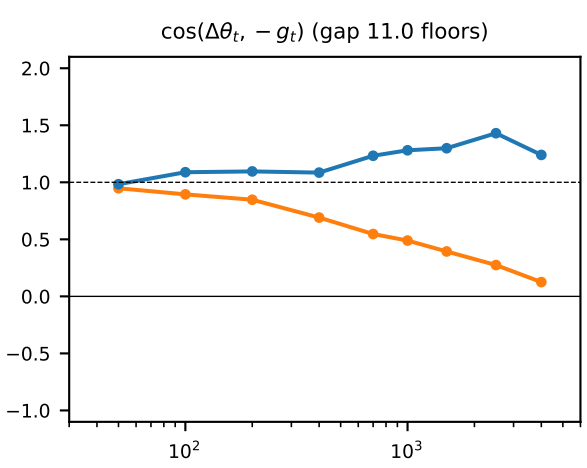


1 = new optimizer (pure B)

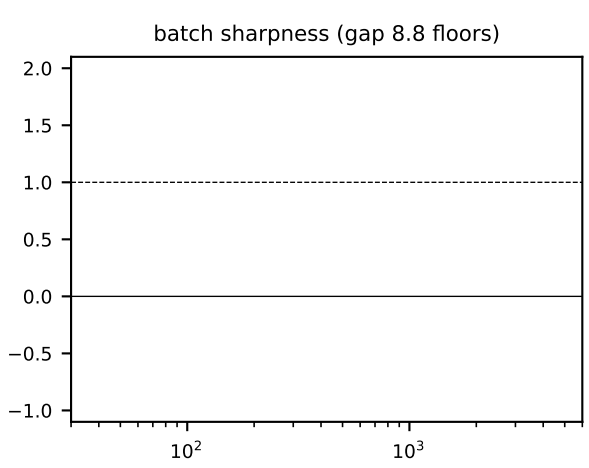
$\|\Delta\theta\|/\|\theta\|$ (gap 114.7 floors)



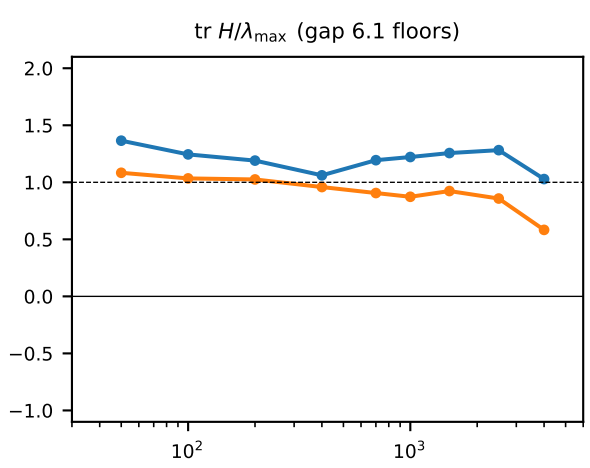
$\cos(\Delta\theta_t, -g_t)$ (gap 11.0 floors)



batch sharpness (gap 8.8 floors)



tr $H/\lambda_{\max}$ (gap 6.1 floors)



shift: 0 = parent (pure A),

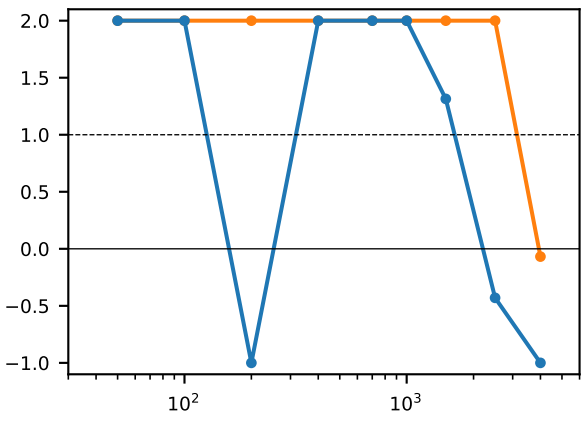
attention on token 1 (gap 0.2 floors)

loss rare - frequent (gap 1.4 floors)

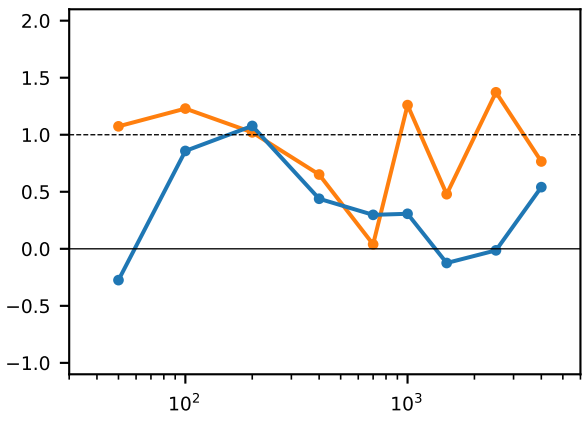
JS to unigram (gap 7.0 floors)

top-1 accuracy (gap 0.2 floors)

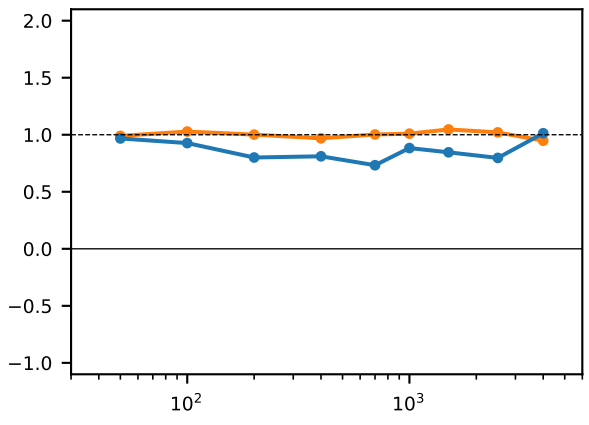
loss, rare targets (gap 3.1 floors)



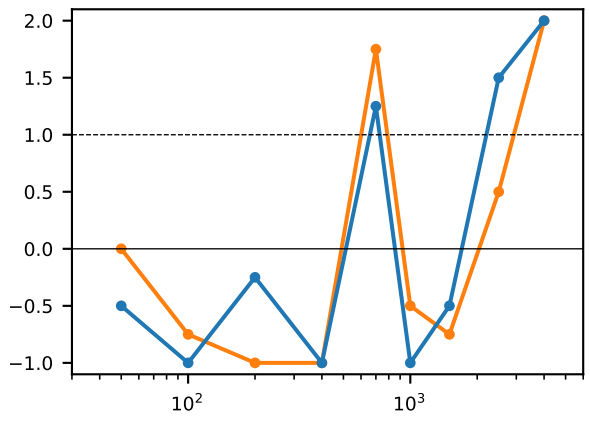
JS to bigram (gap 5.3 floors)



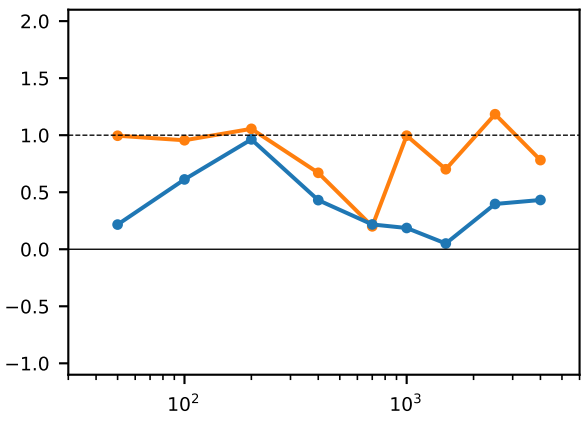
loss, frequent targets (gap 1.5 floors)



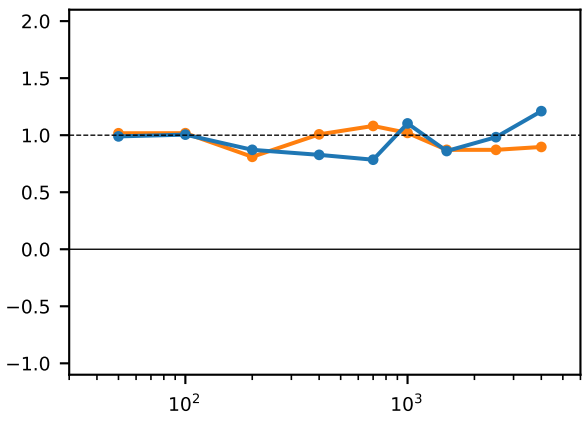
calibration error (ECE) (gap 1.1 floors)



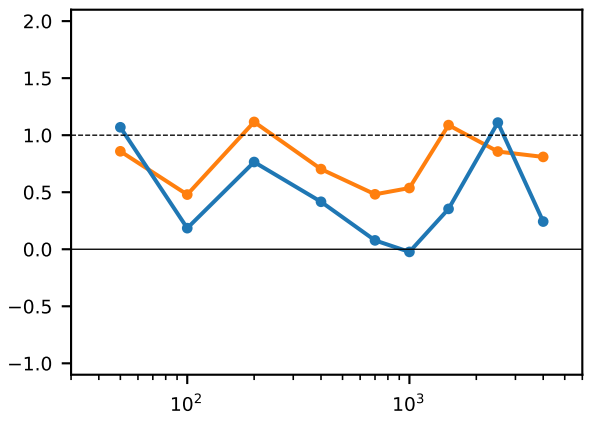
mean top-1 probability (gap 3.1 floors)



prediction entropy (gap 4.3 floors)



val - train loss (gap 0.4 floors)



fork step